BIO 2122 Anatomy & Physiology II

Section 01, Lecture | Fall 2026 | Monday and Wednesday, 6:00 pm to 7:50 pm | Science Hall S-214

Instructor	Dr. Renata Silva, Professor of Biology	**Class meetings**	Monday and Wednesday, 6:00 pm to 7:50 pm
Email	rsilva@sciotocc.edu (best way to reach me)	**Classroom**	Science Hall S-214
Office	Science Hall S-305	**Term**	August 24 to December 16, 2026
Office phone	(614) 555-0142	**Credit hours**	4 (lecture)
Office hours	Monday and Wednesday, 4:30 pm to 5:45 pm, Science Hall S-305, or by appointment	**Prerequisite**	BIO 2121 Anatomy & Physiology I with a grade of C or better
Course site	Scioto Learn: learn.sciotocc.edu/courses/bio2122-fa26	**Section**	01

1. Course Description

BIO 2122 is the second course in the two-semester anatomy and physiology sequence. We study the endocrine system, blood, the heart and blood vessels, the urinary system, fluid, electrolyte and acid-base balance, the respiratory system, the digestive system, nutrition and metabolism, and the reproductive systems. For every system we return to the same question: how does it help keep the internal environment stable? Much of the course is quantitative. You will work problems such as net filtration pressure, cardiac output and arterial blood gas interpretation step by step, the way I work them on the board. The course is designed for students preparing for nursing and allied health programs.

2. Learning Outcomes

By the end of the course you will be able to:

1. Explain how hormones reach and act on their target cells, and describe the feedback control of the major endocrine glands.
2. Describe the composition and functions of blood, including hemostasis and the ABO and Rh blood groups.
3. Trace the conduction system of the heart, relate the ECG and the cardiac cycle to cardiac output, and explain how blood pressure is regulated.
4. Explain how the nephron forms urine through glomerular filtration, tubular reabsorption and tubular secretion, and calculate net filtration pressure.
5. Explain how the body maintains fluid, electrolyte and acid-base balance, and interpret arterial blood gas values.
6. Describe pulmonary ventilation, gas exchange, gas transport and the neural control of respiration.
7. Describe digestion and absorption, nutrient metabolism and energy balance, and the structure and function of the reproductive systems.
8. Use correct anatomical and physiological terminology in writing and on labeled diagrams.

3. Required Materials

1. **Textbook:** *Anatomy & Physiology* (hardcover). Rental copies are available at the Scioto CC Bookstore, North Campus. Chapter numbers in the weekly schedule refer to this text.
2. **Scioto Learn:** lecture slides, practice quizzes, checkpoint quizzes, discussions and grades are posted in our course site.
3. **For every exam:** a No. 2 pencil, your student ID and a basic calculator. Phones may not be used as calculators.

4. How the Course Works

1. **Lectures and slides.** Slides for each lecture are posted on Scioto Learn on the day of the lecture and are numbered in order (L01, L02, L03 and so on). Board work is not posted, so copy worked problems into your notes. The slides set the terminology and notation for this course. On quizzes and exams, use the terms and notation from the slides.
2. **Practice quizzes (ungraded).** Short self-checks on Scioto Learn. A practice quiz usually opens on Wednesday after lecture and closes the following Monday at 11:59 pm. Practice quizzes do not count toward your grade.
3. **Checkpoint quizzes (graded).** Online and individual work. You have one attempt and 45 minutes once you begin. I announce each checkpoint by email when it opens, with the date and time it closes.
4. **Official keys.** I email the official key to the class after each practice quiz and checkpoint quiz closes. If your notes, a classmate's key or any other source disagrees with the official key or the slides, follow the official key and the slides.
5. **Problem sets (ungraded).** Handed out in class for practice with calculations and diagrams. They are not collected. Work each one before the checkpoint quiz on the same material.
6. **Muddiest Point.** Each week, post the idea you found least clear in the Muddiest Point discussion on Scioto Learn. I read every post and begin the next lecture with the most common questions. Posts are optional and ungraded.
7. **Unit exams.** Three unit exams of 100 points each, given in Science Hall S-214 at 6:00 pm on the dates in Section 6. Each exam has multiple-choice questions, labeled diagrams and short worked problems.
8. **Carry-forward sections.** Unit exams are cumulative. Each one includes a carry-forward section, 25% of the exam, on the previous unit. The Unit 1 exam carries forward topics from BIO 2121. I announce the carry-forward topics by email when the previous exam is returned.
9. **Score reports.** Exams are returned in class with a score report that lists your percentage for each topic. Keep your score reports and use them to plan your review for the next exam.
10. **Final exam.** Cumulative. About 40% of the questions cover material taught after the Unit 3 exam.

5. Grading

Component	Weight
Unit 1 Exam	20%
Unit 2 Exam	20%
Unit 3 Exam	20%
Checkpoint quizzes (average of all checkpoints)	10%
Final exam (cumulative)	30%
Practice quizzes and problem sets	Ungraded
Total	**100%**

Grade	Course average (%)
A	93.00 to 100
A-	90.00 to 92.99
B+	87.00 to 89.99
B	83.00 to 86.99
B-	80.00 to 82.99
C+	77.00 to 79.99
C	73.00 to 76.99
C-	70.00 to 72.99
D+	67.00 to 69.99
D	60.00 to 66.99
F	Below 60.00

Course averages are calculated to two decimal places and are not rounded up. Scores are posted in the Scioto Learn gradebook.

6. Exam Schedule and Important Dates

All exams are in Science Hall S-214. Unit exams begin at 6:00 pm.

Date	Event	Details
Monday, August 24	First class	6:00 pm, Science Hall S-214
Monday, September 7	Labor Day	No class
Wednesday, September 16	**Unit 1 Exam, 6:00 pm**	L01 to L06, plus the BIO 2121 carry-forward
Wednesday, October 7	**Unit 2 Exam, 6:00 pm**	L07 to L11, plus the Unit 1 carry-forward
Friday, November 6	Last day to withdraw with a W	See Section 7
Thursday, November 12	**Unit 3 Exam, 6:00 pm**	L12 to L21, plus the Unit 2 carry-forward
Wednesday, November 25	Thanksgiving break	No class
Monday, December 14	Last lecture	Final exam review
Wednesday, December 16	**Final Exam, 6:00 pm to 8:00 pm**	Cumulative, L01 to L27

7. Course Policies

1. **Attendance.** Come to every lecture. If you miss one, review the posted slides and get the board work from a classmate.
2. **Missed exams.** If you cannot attend a unit exam, email me before the exam begins. Make-up exams are given only for documented emergencies and must be taken within one week of the original date. A missed exam without notice is recorded as a zero.
3. **Missed checkpoint quizzes.** Each checkpoint is open for several days, so there are no make-ups. A checkpoint that is not submitted before it closes is recorded as a zero.
4. **Regrade requests.** Submit a regrade request in writing, by email, within one week of the day the exam is returned. Name the question and explain why you would like me to review it.
5. **Communication.** Email is the best way to reach me. Please put "BIO 2122" in the subject line. I reply within one business day, Monday through Friday. Announcements and corrections are sent by email, so check your email at least every other day.
6. **Schedule changes.** The weekly schedule is tentative. Any change is announced in class and by email. If an announcement and this syllabus disagree, follow the most recent announcement from me.
7. **Exam room.** Phones stay in your bag, turned off, for the whole exam. Bring a No. 2 pencil, your student ID and a basic calculator.
8. **Withdrawal.** The last day to withdraw with a W is Friday, November 6, 2026. Please talk with me before you withdraw.

8. Academic Integrity

Checkpoint quizzes and exams are individual work. Until a checkpoint quiz closes, do not give, request or post answers, worked answers or hints for it, whether in person, in group chats, on social media or in shared study materials such as flashcard decks. Study groups are encouraged: discuss concepts at any time, and compare answers once the official key has been emailed. Violations are handled under the Scioto Community College academic integrity policy and may result in a zero for the quiz or exam.

9. Accessibility

If you need accommodations because of a disability, contact the college's disability services office early in the term and give me your accommodation letter. Please allow at least one week before an exam so that testing arrangements can be made with the Testing Center.

10. Weekly Schedule (tentative)

Lecture numbers match the slide files on Scioto Learn. Reading refers to chapters in the textbook; read the chapter before the lecture.

Unit 1: Endocrine System and Blood (L01 to L06)

Date	Lecture	Topics	Reading
Mon, Aug 24	L01	Endocrine system I: hormones, receptors and mechanisms of hormone action	Ch. 16
Wed, Aug 26	L02	Endocrine system II: the hypothalamus and the pituitary gland	Ch. 16
Mon, Aug 31	L03	Endocrine system III: thyroid, parathyroid and adrenal glands; pancreatic islets	Ch. 16
Wed, Sep 2	L04	Blood I: composition of blood, plasma and erythrocytes	Ch. 17
Mon, Sep 7		*Labor Day. No class.*	
Wed, Sep 9	L05	Blood II: hemostasis and blood typing (ABO and Rh)	Ch. 17
Mon, Sep 14	L06	Blood III: leukocytes and platelets	Ch. 17
Wed, Sep 16	**Exam**	**UNIT 1 EXAM, 6:00 pm, Science Hall S-214. L01 to L06, plus the BIO 2121 carry-forward.**	

Unit 2: The Heart, Blood Vessels and Urinary System I (L07 to L11)

Date	Lecture	Topics	Reading
Mon, Sep 21	L07	The heart I: cardiac anatomy, chambers, valves and the pathway of blood through the heart	Ch. 18
Wed, Sep 23	L08	The heart II: cardiac muscle and the conduction system (SA node, AV node, AV bundle, bundle branches, subendocardial conducting network); the ECG	Ch. 18
Mon, Sep 28	L09	The heart III: the cardiac cycle and cardiac output	Ch. 18
Wed, Sep 30	L10	Blood vessels and blood pressure: vessel structure, mean arterial pressure, peripheral resistance, regulation of blood pressure	Ch. 19
Mon, Oct 5	L11	Urinary system I: kidney anatomy, the nephron, glomerular filtration and regulation of GFR	Ch. 25
Wed, Oct 7	**Exam**	**UNIT 2 EXAM, 6:00 pm, Science Hall S-214. L07 to L11, plus the Unit 1 carry-forward.**	

Unit 3: Urinary System II, Fluid, Electrolyte and Acid-Base Balance, Respiratory System (L12 to L21)

Date	Lecture	Topics	Reading
Mon, Oct 12	L12	Urinary system II: tubular reabsorption and secretion	Ch. 25
Wed, Oct 14	L13	Urinary system II: concentrating and diluting urine; renal clearance; micturition	Ch. 25
Mon, Oct 19	L14	Fluid and electrolyte balance: body fluid compartments; water, sodium, potassium, calcium and phosphate balance	Ch. 26
Wed, Oct 21	L15	Acid-base balance I: pH, chemical buffer systems, respiratory and renal regulation	Ch. 26
Mon, Oct 26	L16	Respiratory system I: anatomy of the conducting and respiratory zones	Ch. 22
Wed, Oct 28	L17	Respiratory system II: pulmonary ventilation, lung volumes and capacities	Ch. 22
Mon, Nov 2	L18	Respiratory system III: gas exchange and the transport of oxygen and carbon dioxide	Ch. 22
Wed, Nov 4	L19	Respiratory system IV: control of respiration	Ch. 22
Mon, Nov 9	L20	Acid-base balance II: respiratory and renal compensation	Ch. 26
Wed, Nov 11	L21	Acid-base balance III: acid-base imbalances and arterial blood gases	Ch. 26
Thu, Nov 12	**Exam**	**UNIT 3 EXAM, 6:00 pm, Science Hall S-214. L12 to L21, plus the Unit 2 carry-forward.**	

Unit 4: Digestive System, Metabolism and Reproductive Systems (L22 to L27, tested on the final exam)			
Date	**Lecture**	**Topics**	**Reading**
Wed, Nov 18	L22	Digestive system I: organization of the GI tract; mouth, pharynx, esophagus and stomach	Ch. 23
Mon, Nov 23	L23	Digestive system II: small intestine, liver, gallbladder and pancreas; chemical digestion and absorption	Ch. 23
Wed, Nov 25		*Thanksgiving break. No class.*	
Mon, Nov 30	L24	Nutrition and metabolism I: carbohydrate, lipid and protein metabolism	Ch. 24
Wed, Dec 2	L25	Nutrition and metabolism II: absorptive and postabsorptive states; energy balance and body temperature	Ch. 24
Mon, Dec 7	L26	Reproductive system I: male reproductive system; meiosis and spermatogenesis	Ch. 27
Wed, Dec 9	L27	Reproductive system II: female reproductive system; oogenesis; ovarian and uterine cycles	Ch. 27
Mon, Dec 14	Review	Final exam review. Bring your score reports and your questions.	
Wed, Dec 16	**Exam**	**FINAL EXAM, 6:00 pm to 8:00 pm, Science Hall S-214. Cumulative, L01 to L27.**	

Questions about this syllabus? Email me or come to office hours.